

Viyog: Separating Adversarial and Out-of-Distribution

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Abstract—Safety-critical systems face two different kinds of anomalous input, and they call for opposite responses. Out-of-distribution (OOD) inputs should be abstained on. Adversarial (ADV) attacks should be rejected outright. Most anomaly detectors report only a single score, so they cannot tell the two apart. We present Viyog, a plug-in second stage that needs no retraining and separates OOD from ADV using one simple signal: the dormant-band roughness of the first convolutional activation, i.e. how spatially jagged its quietest channels are. Concretely, this is a magnitude-normalised total variation (the average change between neighbouring pixels). It is read from the forward pass the model already computes, and it stores a tiny $O(C)$ calibration profile, about 0.3 KB for a 64-channel first layer, versus 4.5–40 MB for the feature-distance baselines we evaluate.

We find that, across the four L_∞ attacks we test, adversarial inputs leave more high-frequency residue than natural inputs do in these quiet first-layer channels. Viyog exploits this asymmetry directly: it detects adversarials at held-out AUROC 0.980 (1.0 is perfect, 0.5 is chance), compared with 0.659 for the best confidence/logit score we evaluate. Paired with a logit score, it forms a lightweight three-way ID/OOD/ADV separator reaching balanced accuracy 0.798 on our 17-architecture held-out panel and 0.863 on GTSRB. We also prove that the statistic is invariant to uniform activation rescaling in the zero-offset limit, and give a conditional separation bound for affine layers. Under an adaptive attack suite, a norm-targeted attack achieves 0% evasion, though stronger TV-aware attacks do reduce detector AUROC. The evaluation spans three image datasets, a 17-architecture held-out panel (three of an original 20 architectures were excluded after a data-integrity check, see Sec. V), four attacks, and a synthetic 1D sanity check. Viyog adds no learned parameters and requires no extra forward pass; its statistic costs a measured 2.28% of CNN MACs and 3.9% of inference latency. Viyog is available on PyPI (pip install viyog, pypi.org/project/viyog) and as an interactive live demo on Hugging Face (huggingface.co/spaces/manyagami/viyog).

Index Terms—Out-of-distribution detection, adversarial robustness, feature norms, deep neural networks, safety-critical systems, embedded AI, architecture generalization.

I. Introduction

Deep neural networks (DNNs) now achieve state-of-the-art performance in safety-critical domains: autonomous

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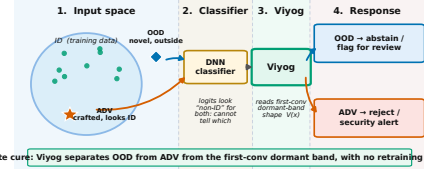


Fig. 1: The core problem: OOD and ADV inputs require opposite responses, yet both fool the classifier. (1) Input space: an OOD (out-of-distribution) input lies outside the training cluster; an adversarial (ADV) input is deliberately crafted to sit inside the cluster while forcing a wrong prediction. (2) Classifier: both can produce logits that the evaluated output-side detectors do not reliably distinguish. (3) Viyog: reads the dormant-band roughness $V(x)$ of the first convolutional activation map (a gradient-free statistic with $O(C)$ stored state) and separates the two. (4) Response: OOD $\rightarrow$ abstain / flag for review; ADV $\rightarrow$ reject / security alert. Conflating them wastes adversarial defences on benign inputs or silently accepts attacks.

driving, medical imaging, embedded vision. But a deployed system also needs to detect abnormal inputs and respond to them appropriately before it acts on a prediction. Standard classification pipelines do not do this [1].

Two kinds of anomalous input require opposite responses (Fig. 1). OOD inputs fall outside the training distribution, so the right response is to abstain. Adversarial (ADV) inputs are crafted to force a misclassification, so the right response is to reject them outright. Treating the two the same way is costly either way: it wastes a defence on a benign input, or it lets an attack through silently. Most lightweight baselines evaluated here do not draw this distinction at all. We investigate a signal that does: one found in the network’s first layer.

We identify an asymmetry in first-layer activation statistics that is consistent across the evaluated models and can be read with low measured overhead. For the evaluated digital attacks, the first convolutional or projection layer contains a rough, broadband-like residue in quiet (“dormant”) channels, while the tested ID and OOD inputs generally leave those channels smoother. Prior work also reports early-layer and frequency-domain effects of adversarial perturbations [2], [3]. This is an empirical regularity of our threat model, not a property guaranteed for every adversarial example. An earlier, weaker form of Viyog exploited this asymmetry with the raw first-layer L_∞ norm (the L_∞ -form): adversarials fall below the ID mean and OOD at or above it. But the L_∞ -form is regime-bounded: strong only for far-OOD (inputs far from the

training distribution, e.g. a different dataset) and near-chance for near-OOD (visually similar shifts) under a strict cross-model protocol. The primary signal is instead the relative roughness of the dormant channels: a magnitude-normalised total-variation (TV) statistic (Viyog, Sec. IV) that measures shape (spatial roughness), not magnitude. In the zero-offset limit it is a 0-homogeneous ratio, so uniformly rescaling a feature map leaves it unchanged. This fact does not cover general norm-preserving perturbations, which may change spatial roughness. It is complementary to logit OOD scores and yields three-way ID/OOD/ADV separation (Sec. VI); the phenomenon persists across CNN and transformer architectures and across 32×32 and 224×224 inputs. We ground this in a formal Proposition (Section III) that gives a deterministic bound for affine pre-activations under explicit smoothness, residue-roughness, and residue-size conditions.

This paper makes five core contributions:

C1: The Viyog scoring statistic. A gradient-free first-layer statistic (the magnitude-normalised total variation of the quietest dormant channels) that separates ADV from OOD where the raw L_∞ norm cannot (mechanism confirmed on real ResNet-50 features, Fig. 2b). The continuous statistic $V(x)$ is itself the graded degree-of-anomaly readout; the monotone routing squash is retained only to shape the binary operating curve, not as the severity scale. The raw L_∞ score is kept as the motivating baseline.

C2: Theory for the shape statistic. A formal Proposition (Sec. III-B) proving that the dormant-band roughness statistic is invariant to uniform rescaling at zero stabilisation offset and bounding its change under the exact affine residue $e_c = w_c * \delta$. The separation result holds only when the stated smoothness and rough-residue conditions hold for the affine pre-activation; post-activation ReLU/GELU behavior is evaluated empirically rather than covered by the bound. The ID/OOD/ADV ordering is therefore a conditional consequence, not a worst-case guarantee; a general-affine operator-norm envelope (Appendix A) covers the raw- L_∞ baseline.

C3: Regime characterisation and a complementary dormant-band detector. Delimiting where the global L_∞ mechanism holds and recovering the lost near-OOO/strong-attack regime with Viyog, we obtain a low-overhead three-way ID/OOD/ADV separation: across the 17-architecture held-out panel and two datasets Viyog is complementary to logit OOD scores, and a lightweight supervised head over the full detector panel reaches balanced accuracy 0.798 (CIFAR-100, 17 archs) / 0.863 (GTSRB, 6 archs), the logit+Viyog pair alone 0.758/0.768, with $O(C)$ added state (about 0.3 KB at 64 channels), validated on ConvNeXtV2, ViT-B/16, and Swin-T at 224×224 .

C4: Comprehensive empirical validation. Experiments across the dataset suite in Table II and four L_∞ attacks. Under the far-OOO far-OOO single-direction protocol the raw L_∞ score outperforms ten established baselines by up to +24.9 AUROC points. The stricter directionless re-evaluation (which disqualifies any score whose high/low

TABLE I: Representative anomaly detectors and their OOD-ADV separation capability in our evaluation. $\dagger$ Viyog scores both T2 and T3 above chance while remaining gradient-free and state-minimal.

Family	Examples	Score	Sep. OOD/ADV?	Cost
Confidence	MSP, Energy, ODIN	logit	$\times$ both high conf.	$O(1)$
Distance	Maha, KNN, ViM	feat. dist.	$\times$ ADV inside manifold	$O(D^2) - O(ND)$
Uncertainty	MCD	post. var.	$\times$ single score	$30 \times$ fwd
Joint	UNICAD [7]	proto./DAE	partial	trained modules
Viyog †	this work	shape V	$\checkmark$ T20.980 / T30.844	$O(C)$

meaning flips between models) over the 17-architecture held-out panel is reported in Table III.

C5: Adaptive attack analysis. In the evaluated attack suite, the norm-targeted attack achieves 0% evasion, a both-aware adaptive attacker pays an 8% success cost, and the stochastic Expectation-over-Transformation (EOT) defence, which randomises which dormant channels are read so the attacker cannot lock onto a fixed target, restores the panel to 0.82 (Fig. 13).

II. Background and Related Work

A. OOD Detection Methods

Table I categorises representative prior work; surveys are in [4], [5]. Most evaluated detectors assign a single anomaly score rather than an explicit OOD-versus-ADV decision. Szyk et al. [6] show that rankings are protocol-sensitive, motivating our multi-seed evaluation (Sec. V).

B. Adversarial Attacks, Defences, and Early-Layer Effects

Adversarial attacks [8] solve $\min_\delta \|\delta\|_p$ s.t. $f(x+\delta, \theta) \neq y$. Common attacks include FGSM [9], PGD [10] (both L_∞), and L_2 -constrained CW [11] and DeepFool [12]. A fundamental tension exists between robustness and accuracy [13], [14].

Defences against adversarial attacks fall into several families: input preprocessing [15], feature denoising [16], generative purification [17]–[20], and adversarial training [10], [14]. Ensemble and Jacobian-based methods provide additional robustness but at increased inference cost. These defences do not generally target the OOD/ADV disambiguation problem; they assume the threat class is known in advance, precisely the assumption Viyog is designed to relax.

Early-layer effects are well documented: the first convolutional layer is disproportionately critical for robustness [3] and acts as an implicit noise filter [2] (the front-neuron concept formalised here); pruning low-activation neurons improves robustness by forcing reliance on high-magnitude responses [21], and restoring suppressed activations [22] presupposes the suppression Viyog detects. Filter-statistic [23] and frequency-domain [24], [25] analyses likewise distinguish adversarial perturbations from natural shift at early layers, but none exploits the OOD/ADV asymmetry. These observations motivate the conditional pre-activation proposition in Section III; the result formalises sufficient roughness-separation conditions rather than universal activation suppression.

C. Joint OOD and Adversarial Detection

A small body of work attempts the OOD–ADV separation directly, but at a cost Viyog avoids. Zhang et al. [26] propose an OOD detector based on adversarial gradient-attribution paths. It is not an OOD-versus-ADV router and requires gradient computation at inference time. Feng et al. [27] propose a unified open-world detection perspective but require joint training with anomaly labels. Walkowiak et al. [28] combine outlieriness scores and feature extraction, while UNICAD [7] unifies attack detection and novel-class identification in a single framework. Detection methods based on feature statistics [23] and disentangled representations have also been proposed. Unlike gradient-attribution or joint-training approaches, Viyog operates in a single gradient-free forward pass and serves as a plug-in second stage atop any existing non-ID detector, without architectural modification. The reported held-out protocol uses a small labelled ID/OD/ADV calibration split to fix score polarity and operating points; the optional three-way head uses the same split (Sec. VI). This efficiency advantage is quantified in the system analysis that follows.

D. System Efficiency and Deployment Motivation

For embedded systems inference cost is a primary constraint [1], yet prior detectors are expensive (ODIN and gradient-norm backward passes, a full Mahalanobis covariance, MCD’s 30 forward passes ($\sim 30\times$ a single inference), diffusion denoising). Viyog adds zero parameters, no gradient computation, and uses a first-layer activation hook plus a spatial reduction (read as a dormant-band shape statistic); the pipeline below motivates a candidate system integration.

In the Viyog two-stage pipeline (Fig. 2a), a standard inference pass is followed by a first-stage non-ID detector, and Viyog attempts to partition flagged inputs into OOD and ADV using the first-layer activation map and the dormant-band roughness statistic V . This second stage provides a candidate signal for choosing different responses; its errors and end-to-end recall remain relevant in deployment (Sec. IX).

III. Theoretical Motivation and Analysis

We formalise a sufficient condition under which a rough additive residue raises the dormant-band statistic at an affine pre-activation (Prop. 1). The result proves uniform-rescaling invariance in the zero-offset limit and conditional separation; it is not a worst-case guarantee for arbitrary attacks or nonlinear post-activations. The argument is rooted in the convolutional nature of the first layer shared by all architectures evaluated: strided convolutions in ResNet [31], DenseNet [32], and ConvNeXtV2 [33]; and patch-embedding projections (which are mathematically equivalent to strided convolutions) in ViT [34] and Swin-T [35]. The insight that high-frequency and norm-altering perturbations are concentrated at early layers is supported by prior frequency-domain analyses [15], [24], [25].

A. Why the First Layer?

Natural-image smoothness (empirical premise A1). Natural image power spectra decay as $P(f) \propto f^{-2}$: low-frequency components (slow gradients, textures) dominate; rapid pixel-to-pixel oscillations are negligible. Each first-layer channel c produces a feature map $a_c \in \mathbb{R}^{H \times W}$, the response of filter w_c to input x . For the tested natural inputs, a_c is typically smooth: its Total Variation $\text{TV}(a_c)$ (mean pixel-to-pixel jump, defined formally in Eq. (1)) is small relative to the channel’s mean magnitude $|a_c|$. Proposition 1 makes the required per-channel inequality explicit; natural OOD inputs that violate it are a failure mode rather than an exception to the premise.

Rough adversarial residue (empirical premise A2). Gradient attacks add a perturbation δ to raise the loss. At an affine first layer, each filter w_c maps it to the exact additive residue $e_c = w_c * \delta$. The four evaluated L_∞ attacks usually produce high relative TV in the selected channels. An L_∞ constraint alone does not impose this spectral property, however; a smooth or detector-aware perturbation can violate A2.

Why a normalised ratio, not a raw norm (0-homogeneity). A raw L_∞ norm measures magnitude, not shape. OOD images can produce larger or smaller first-layer activations than ID images, so $\|a_c\|_\infty$ is direction-inconsistent; an attacker can further append a norm-preserving penalty to keep $\|a'_c\|_\infty \approx \|a_c\|_\infty$ while misclassifying. The normalised ratio $\text{TV}(a_c) = \text{TV}(a_c)/(|a_c| + \epsilon)$ (small $\epsilon > 0$ for stability) addresses magnitude variation: TV and $|\cdot|$ are both 1-homogeneous, so their ratio is 0-homogeneous when the stabilising offset is zero (P1 below). Thus uniform feature-map rescaling does not change the idealised ratio. Preserving a norm while changing spatial structure is not uniform rescaling and can change TV; the adaptive experiments measure, rather than rule out, that case.

B. The Dormant-Band Roughness Statistic

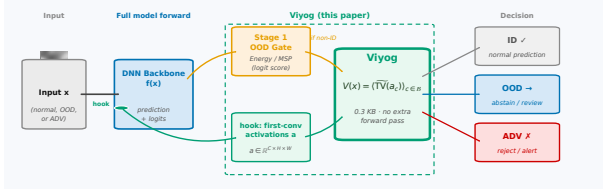
The statistic. Let $a = f_0(x) = W * x + b \in \mathbb{R}^{C \times H \times W}$ be the first-layer activation map (C channels, spatial size $H \times W$), with $a_c \in \mathbb{R}^{H \times W}$ the feature map of channel c , filter w_c , and bias b_c . Define:

$$V(x) = \frac{1}{|\mathcal{B}|} \sum_{c \in \mathcal{B}} \widetilde{\text{TV}}(a_c), \quad \widetilde{\text{TV}}(a_c) = \frac{\text{TV}(a_c)}{|a_c| + \epsilon}, \quad (1)$$

$$\text{TV}(a_c) = \frac{1}{2} (|\Delta_h a_c| + |\Delta_w a_c|),$$

where $\Delta_h a_c[i, j] = a_c[i+1, j] - a_c[i, j]$ and $\Delta_w a_c[i, j] = a_c[i, j+1] - a_c[i, j]$ are pixel differences, $\bar{u} = \frac{1}{HW} \sum_{i,j} u_{ij}$ is the spatial mean, and $\epsilon = 10^{-6}$ prevents division by zero. The dormant band is $\mathcal{B} = \text{bottom-}p\% \text{ of alive channels}$, where $\mathcal{A} = \{c : \mathbb{E}_{\text{ID}} |a_c| > \kappa\}$ ($\kappa = 10^{-4}$) excludes permanently silent channels.

For the analysis below, a denotes the affine pre-activation. A small denominator can make the dormant band sensitive to roughness, but the normalisation also makes dead-channel exclusion essential. Implementations that hook a post-nonlinearity are evaluated empirically;



(a) Candidate Viyog two-stage pipeline. A normal forward pass drives a first-stage OOD gate (logit-based scores such as Energy [29] or MSP (Maximum Softmax Probability [30])) and exposes the first-conv activation map $a = f_0(x)$ through a hook. Flagged inputs enter Viyog, which computes $V(x)$, the dormant-band roughness (total-variation shape statistic) of a , and routes to ID (pass), OOD (abstain / flag), or ADV (reject / alert), adding $O(C)$ state (about 0.3KB at 64 channels), no extra forward pass, and the measured spatial-reduction cost reported in Sec. VI.

they are not covered by the affine proposition. Fig. 2b confirms this on real ResNet-50 features: $V(x)$ separates by type of signal (natural vs. attack), not by size of activation.

Proposition 1 (Conditional relative-roughness bound for an affine channel). Fix a channel $c \in \mathcal{B}$ of the affine pre-activation and write $a_c = w_c * x + b_c$, $\nu_c := \overline{|a_c|} > 0$. For $x' = x + \delta$, affineness gives $a'_c = a_c + e_c$ exactly, where $e_c = w_c * \delta$. Define $r_c := \overline{|e_c|} / \nu_c$, $\rho_c := \text{TV}(e_c) / |e_c|$ when $\overline{|e_c|} > 0$, and $\eta_c := \epsilon / \nu_c$. Assume:

- (A1) Natural smoothness: $\text{TV}(a_c) \leq \beta_c \nu_c$; the same β_c upper-bounds the natural ID/OOD comparison class.
- (A2) Rough residue: $\rho_c > 0$ for the perturbation under consideration.
- (A3) Nonzero residue: $r_c > 0$.

Then:

- (P1) Uniform-rescaling identity: for $\lambda_c > 0$,

$$\widetilde{\text{TV}}(\lambda_c a_c) = \frac{\text{TV}(a_c)}{\nu_c + \epsilon / \lambda_c}.$$

Thus the statistic is exactly scale-invariant at $\epsilon = 0$, and only approximately so when $\epsilon / (\lambda_c \nu_c)$ is negligible.

- (P2) Deterministic two-sided bound:

$$\frac{[\rho_c r_c - \beta_c]_+}{1 + r_c + \eta_c} \leq \widetilde{\text{TV}}(a'_c) \leq \frac{\beta_c + \rho_c r_c}{|1 - r_c| + \eta_c},$$

where $[z]_+ = \max(z, 0)$. The upper bound becomes loose near $r_c = 1$, where cancellation can make the normalising denominator small.

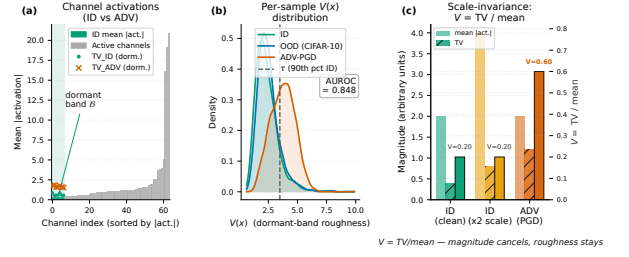
- (P3) Conditional separation: if

$$\rho_c r_c > \beta_c(2 + r_c + \eta_c)$$

for every $c \in \mathcal{B}$, then the perturbed pre-activation has larger V than any natural comparison satisfying A1. The per-channel margin is at least

$$\frac{\rho_c r_c - \beta_c(2 + r_c + \eta_c)}{1 + r_c + \eta_c} > 0.$$

P3 is a sufficient condition, not a guarantee that A1–A3 hold for an unseen natural distribution or attack. ReLU



(b) The Viyog mechanism confirmed on real ResNet-50 features (CIFAR-100, $n=2000$ samples per class). (a) Dormant-band channels (bottom 10% alive, green) have the smallest denominators, amplifying injected roughness. (b) ID and OOD co-cluster at low $V(x)$; ADV-PGD spikes high (AUROC shown; dashed line = calibrated threshold). (c) Uniform-rescaling check (P1): doubling activation magnitude leaves V approximately unchanged at the chosen ϵ ; the adversarial example has the elevated roughness ratio in this experiment.

is 1-Lipschitz, but that fact alone does not preserve a bound on a normalised ratio; GELU is neither globally monotone nor 1-Lipschitz. Consequently P2–P3 are not asserted for nonlinear post-activation maps. Appendix A instead bounds the local GELU Taylor remainder, and Section VI reports the measured behavior.

Proof sketch (full proof in Appendix A). P1. Substitute the absolute 1-homogeneity of TV and $|\cdot|$ and divide numerator and denominator by λ_c .

P2. TV and the spatial L_1 norm satisfy triangle and reverse-triangle inequalities:

$$\begin{aligned} \text{TV}(a_c + e_c) &\leq \text{TV}(a_c) + \text{TV}(e_c), \\ \text{TV}(a_c + e_c) &\geq \text{TV}(e_c) - \text{TV}(a_c). \end{aligned}$$

Also, $\nu_c |1 - r_c| \leq \overline{|a_c + e_c|} \leq \nu_c (1 + r_c)$. Substituting $\text{TV}(e_c) = \rho_c r_c \nu_c$, adding ϵ to the denominator, and taking the nonnegative part of the lower numerator gives P2.

P3. Any natural comparison satisfying A1 has normalised TV at most β_c . Subtracting this from P2's lower bound gives the displayed margin; averaging positive per-channel margins over the fixed band gives the result. $\square$

In plain words. When a natural feature map is smooth and the additive affine residue is sufficiently large and rough, the bound predicts a higher V . The premise can fail for naturally rough OOD inputs or attacks that optimise the spatial statistic, so the prediction must be checked empirically under the intended threat model.

IV. The Viyog Method

A. Step-by-Step Derivation and Motivation

The first design choice, reading at the first convolutional layer, follows directly from the theory of Section III; the remaining three (restricting to the dormant band, using a total-variation shape statistic, and held-out calibration) are developed in the subsections that follow:

First convolutional layer. The first convolutional layer has an exact affine perturbation relation before its nonlinearity [2], [3], and, crucially, the adversarial residue has

not yet been transformed or pooled. The shape statistic V peaks at the first conv (mean T2 0.965) and decays with depth in the six-model, two-attack depth sweep; it is therefore the lowest-cost and highest-AUROC layer in that experiment (Fig. 9).

Given this shape-based score, the routing rule is straightforward.

B. Decision Rule and Downstream Routing

Given a pool of non-ID inputs flagged by a first-stage detector, Viyog applies a second-stage partition:

- $Viyog(x) > \tau$: ADV, apply adversarial defence, rejection, or security alert.
- $Viyog(x) \leq \tau$: OOD, apply abstention, fallback, or incremental-learning trigger.

A magnitude threshold allows precision–recall trade-offs. In the reported protocol, score polarity and threshold τ are fixed from a labelled ID/OOD/ADV calibration split and then applied once to disjoint test data (Alg. 1); the end-to-end gate uses the same held-out protocol (Sec. IX-A).

C. Dormant-Band Shape Statistic (Viyog)

The primary statistic $V(x) = Viyog(x)$ uses the dormant band of Eq. (1). Restricting it to active channels is essential: up to 25/64 first-conv channels are permanently silent (e.g. DenseNet-121) and would otherwise force a degenerate score. Across the 37-statistic battery V tops both the ID-vs-ADV and OOD-vs-ADV rankings (Fig. 3).

D. Algorithm and Complexity

Fig. 4 provides a step-by-step visual walkthrough of the full Viyog pipeline with activation-map images at each stage.

Algorithm 1 summarises the Viyog procedure: held-out calibration computes the per-channel ID profile, dormant band $\mathcal{B}$, polarity, and operating point; inference then adds only $\mathcal{O}(|\mathcal{B}| h_1 w_1)$ operations and $\mathcal{O}(C)$ stored state ($\mathcal{O}(C)$; about 0.3 KB at 64 channels), measured as 2.28% of CNN MACs and 3.9% of H200 inference latency, with four to five orders of magnitude less stored state than the evaluated feature-distance detectors. We now validate it against eleven baselines across twenty model–dataset configurations.

V. Experimental Setup

A. Datasets and Models

All experiments use two primary ID datasets: CIFAR-100 [36] (224×224 , the main benchmark) and the safety-critical GTSRB [37] (224×224). Per-dataset OOD sets are drawn from SVHN [38], iSUN, LSUN, and Places365; MNIST serves as a supplementary near-OOD probe for DenseNet.

The model panel comprises 17 architectures spanning every major design family: ResNet-50/101/152 [31],

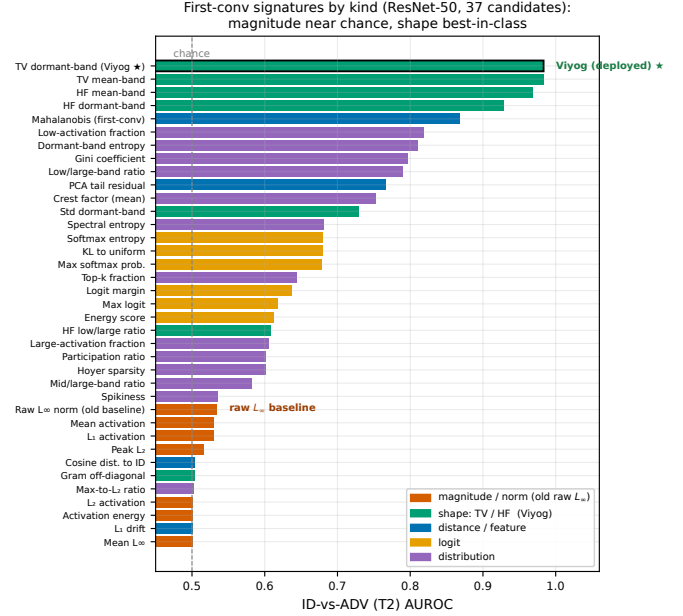


Fig. 3: First-conv signature ranking by ID-vs-ADV (T2) AUROC on ResNet-50 (37 candidates), coloured by family. Magnitude/norm (orange), including the raw L_∞ baseline, sits near chance (0.50–0.53); shape (green, TV/HF dormant-band) is best-in-class (≈ 0.98) with Viyog first. Distance (blue), logit (yellow), and distributional (purple) statistics fall in between: shape, not magnitude, separates ADV from natural inputs here.

DenseNet-121/161/169/201 [32], ConvNeXtV2 [33], ViT-B/16 [34], Swin-T [35], MobileNetV3-L, MobileNetV4-M, EfficientNet-Lite0, EfficientNetV2-L, EfficientViT-B1, FastViT-SA12, and EdgeNeXt-Small, all fine-tuned from ImageNet pretraining in PyTorch [39] without adversarial augmentation. (ResNet-18/34 and MobileOne-S1 were excluded: their archived first-layer features are identically zero, Table VI.) Training-seed confidence intervals (≤ 3 seeds per architecture) are reported throughout; per-architecture std on T2 is ≤ 0.004 (Fig. 7).

B. Attacks, Baselines, and Metrics

Four L_∞ attacks span the threat model: FGSM [9], BIM [40], PGD [10], and APGD-CE [41] via TorchAttacks [42] at $\epsilon=8/255$. Eleven pytorch-ood [43] baselines, MSP, Entropy, MaxLogit, Energy, GEN, ODIN ($T=1000$), MCD (30 passes), Mahalanobis, KNN, OpenMax, and KL-Matching, are compared on AUROC, FPR@95TPR (false-positive rate at 95% true-positive rate), and AUTC (area under the threshold curve, critical for deployment where post-hoc tuning is infeasible). We evaluate three tasks: T1 ID-vs-OOD, T2 ID-vs-ADV (the security task), and T3 OOD-vs-ADV (the second-stage routing task).

Held-out calibration protocol. For the main CIFAR-100 evaluation, each pool is partitioned deterministically into 20% calibration and 80% test data (split seed 0). Calibration data alone determine the alive/dormant channels (absolute threshold $p_c > 10^{-4}$), KL-Matching references, score polarity, and operating points. Polarity

How Viyog works: one forward pass, no extra parameters

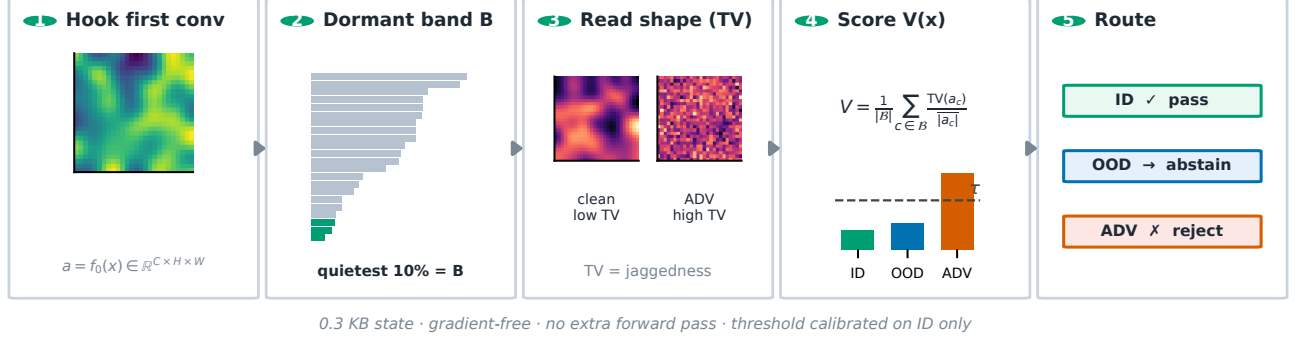


Fig. 4: How Viyog works, in five steps (one forward pass, no extra parameters). (1) Hook the first-conv activation map $a = f_0(x)$. (2) Rank channels by mean ID activation; the quietest 10% form the dormant band $\mathcal{B}$ (green). (3) Read spatial roughness (total variation) on $\mathcal{B}$: clean channels are smooth (low TV), adversarial channels are jagged (high TV). (4) Score $V(x)$, the mean normalized TV over $\mathcal{B}$: the calibration split fixes polarity and threshold τ . (5) Route the input: ID passes, OOD abstains, ADV is rejected. Full method and routing rule in Sec. IV (Alg. 1).

Algorithm 1 Viyog: Dormant-band shape statistic for OOD-vs-ADV separation

Require: Model f with first affine layer f_0 ; labelled calibration sets $\mathcal{D}_{\text{ID}}, \mathcal{D}_{\text{OOD}}, \mathcal{D}_{\text{ADV}}$; band fraction $\rho=0.1$; test sample x

- 1: Calibration (once per model):
- 2: $p_c \leftarrow \frac{1}{|\mathcal{D}_{\text{ID}}|} \sum_{x \in \mathcal{D}_{\text{ID}}} |f_0(x)_c|$ $\triangleright$ per-channel ID mean activation
- 3: $\mathcal{A} \leftarrow \{c : p_c > 10^{-4}\}$ $\triangleright$ drop permanently-dead filters
- 4: $\mathcal{B} \leftarrow$ bottom- ρ of $\mathcal{A}$ by p_c $\triangleright$ dormant band
- 5: $(d, \tau) \leftarrow$ fit polarity $d \in \{-1, +1\}$ and operating point on labelled calibration scores $\triangleright$ then freeze
- 6: Inference (per sample, single forward pass):
- 7: $a \leftarrow f_0(x)$ $\triangleright$ hook; no extra pass, no gradients
- 8: $s \leftarrow \frac{1}{|\mathcal{B}|} \sum_{c \in \mathcal{B}} \widetilde{\text{TV}}(a_c)$ $\triangleright$ dorm-band roughness, Eq. (1)
- 9: if $ds > \tau$ then
- 10: return ADV (reject / defend) $\triangleright$ high-frequency dorm-band residue
- 11: else
- 12: return OOD (abstain / incremental learn)
- 13: end if

Optional 3-way classifier: fit a Linear Discriminant Analysis (LDA) on the labelled calibration split over the feature vector $[\text{logit score}, V(x), \dots]$. This adds $<0.1\text{KB}$ of state and returns a three-class ID/OOD/ADV decision (Sec. VI).

and operating points may use labelled ID/OOD/ADV calibration samples; they are then frozen and evaluated once on the disjoint test split. Reported AUROCs use the calibration-selected polarity, not a test-set oracle or $\max(\text{AUROC}, 1 - \text{AUROC})$. A $\times$ in Table III indicates that the raw score’s high/low direction varies across architectures; it is a diagnostic rather than a post-hoc test orientation rule [6]. Table II studies marked $\dagger$ predate the

TABLE II: Experimental coverage. Main detection uses a deterministic 20% calibration / 80% test split. $\dagger$ predates the data-integrity fix (Table VI), not rerun on the 17-arch panel.

Experiment	Coverage	Measures
Main T1/T2/T3	CIFAR-100, 17 usable archs, 20/80 held-out split	AUROC, FPR@95, recall
Measured cost	CIFAR-100, H200 (CUDA events)	latency, mem, energy
Layer-depth sweep	6 archs $\times$ 2 attacks $\times$ all depths	T2/T3 vs. depth
Adaptive A0-A3, EOT $\dagger$	20 archs; norm/TV-aware + stochastic	evasion %, AUROC
Strong adaptive A4	6 archs (C100+GTSRB), held-out basis	detector floor
Training-seed CIs $\dagger$	20 archs $\times$ ≤ 3 seeds	T2/T3 std
Non-vision sanity	1D signals, FGSM/PGD	T2 AUROC

exclusion above.

Experimental coverage. Table II lists the experiments. The main detection sweep uses the held-out protocol above; specialised studies use the protocols stated in their sections. Beyond the main detection sweep we add a layer-depth sweep (Fig. 9), a measured on-silicon cost benchmark against all eleven baselines (Figs. 11, 12), an adaptive ladder with an EOT stochastic-band defence (Sec. VII, Fig. 13), training-seed CIs (Fig. 7), and a 1D non-vision sanity check (Appendix B). The adaptive ladder and training-seed CIs were run on the original 20-architecture panel before the MobileOne-S1/ ResNet-18/ResNet-34 data-integrity issue was identified and have not been rerun against the corrected 17-architecture panel for this submission (Table VI).

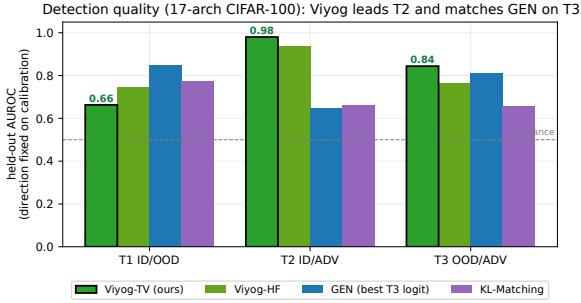


Fig. 5: Detection quality across the 17 CIFAR-100 architectures, by task and method family. Viyog leads on both security tasks: T2 0.980 (ID-vs-ADV) and T3 0.844 (OOD-vs-ADV), while logit detectors win the T1 ID/OOD task (0.86). Tasks are scored directionlessly. At a strict 5%-FPR point mean ADV recall is 0.957 (median 0.964); Sec. IX-A reports an end-to-end cascade analysis.

VI. Results and Discussion

A. Detection Quality Across 17 Architectures

With the evaluation protocol of Sec. V fixed, we first report raw detection quality across all 17 usable backbones.

Under the held-out 17-architecture protocol (Table III, Fig. 5), Viyog leads on the security task T2 (0.980) and T3 (0.844) with $O(C)$ state. The five evaluated confidence/logit scores (Energy, MSP, MaxLogit, Entropy, GEN) score at most 0.647 on T2; direction-consistent KL-Matching reaches 0.659. A three-way supervised head reaches 0.798 balanced accuracy (Fig. 6). The T3 signal is above chance across the evaluated 17 usable architectures, four attacks, and three datasets (Fig. 8); GTSRB provides a second-ID-dataset check.

The failure modes are complementary: logit scores collapse on T2 (blind to high-confidence adversarials), while feature-distance detectors (Mahalanobis, KNN) trail on T3, having pulled perturbations back toward the ID manifold at deep layers [44]. Among the compared single statistics, the first-conv shape statistic separates both (Sec. III-A).

A limited training-seed check (at most three seeds per architecture) shows small T2 variability in this panel (Fig. 7); broader training and data-split replication remains future work.

B. Near-OOD Difficulty and Shape-Depth Sweep

We break the T3 result by OOD difficulty and compare the shape read across depth. In the evaluated splits, ID/OOD channels are generally smoother than their adversarial counterparts (Fig. 15).

Harder near-OOD and texture splits. Because near-distribution or texture-biased OOD may themselves perturb early activations, we break the 17-architecture OOD-vs-ADV (T3) result by difficulty. Viyog reaches far-OOD (0.96) and degrades gracefully but stays above chance on near-OOD (0.74 mean, up to 0.95) and texture-OOD (0.67); near-OOD is indeed the harder regime. A retrospective high-frequency diagnostic reaches 0.92 on texture-

TABLE III: Held-out evaluation across the 17-architecture CIFAR-100 panel (of an original 20; MobileOne-S1, ResNet-18, and ResNet-34 excluded, Sec. V). Setup: T1/T2/T3 are defined in Sec. V. Metric: mean AUROC / FPR@95TPR (%). $\times$ = direction-inconsistent; best AUROC per column in bold; shaded row = primary detector; state memory in last column. Highlights: five of six logit baselines are direction-inconsistent on T2; KL-Matching is consistent and reaches 0.659, vs. at most 0.647 for the rest. Raw Viyog L_∞ is itself direction-inconsistent, whereas Viyog has the highest mean T2 (0.980) and T3 (0.844) among compared methods with $O(C)$ state. Impact: the evaluated signal is carried by first-conv shape, not magnitude or logits.

Method	AUROC / FPR@95TPR (%)			State
	T1 ID/OOD	T2 ID/ADV	T3 OOD/ADV	
Logit baselines (pytorch-ood)				
Energy	0.863 / 52.1	0.600 $\times$ / 98.0	0.790 / 80.5	4B
MSP	0.837 / 61.9	0.621 $\times$ / 98.4	0.796 / 79.1	4B
MaxLogit	0.864 / 52.5	0.599 $\times$ / 98.6	0.794 / 79.5	4B
Entropy	0.848 / 62.1	0.627 $\times$ / 98.7	0.803 / 77.0	4B
GEN	0.848 / 63.4	0.647 $\times$ / 98.6	0.812 / 75.6	4B
KL-Match	0.775 / 82.5	0.659 / 79.7	0.655 / 83.7	4B
First-layer statistics (ours)				
FirstConv- L_∞ (baseline)	0.562 $\times$ / 98.1	0.515 $\times$ / 87.6	0.577 / 75.0	8B
Viyog-HF (high-freq. dorm.)	0.745 / 83.3	0.939 / 25.9	0.763 / 59.4	$O(C)$
Viyog (primary)	0.663 / 96.5	0.980 / 4.9	0.844 / 37.4	$O(C)$

OOD where the total-variation read is weakest (Fig. 10). Because the displayed panel selects the better statistic per architecture using evaluation performance, it is an oracle upper envelope, not an operational combination; a selection rule must be fixed on disjoint calibration data. In the six-architecture, two-attack depth sweep, the shape read peaks at depth 0 (mean T2 0.965, T3 0.918; the higher T3 than the 17-architecture main mean 0.844 reflects the easier subset) and decays with depth; this experiment does not establish optimality for unseen architectures or attacks (Fig. 9).

C. Computational Efficiency

We next compare the accuracy and cost of the evaluated detectors.

Figs. 11–12 and Table IV quantify the cost: Viyog uses $O(C)$ state (about 0.3KB at 64 channels) and 2.28% of CNN MACs with no extra forward pass. Its measured incremental detector latency is 3.9% of full inference on the H200; edge-energy values below are roofline projections, not on-device measurements. The standalone raw-input score-path timings in Fig. 11 are not deployed speedup measurements.

Together the AUROC, AUTC, latency, and memory results place Viyog favorably among the evaluated methods. We next stress-test the detector against an adaptive adversary.

VII. Adaptive Attack Evaluation

A rigorous adversarial evaluation must include an adaptive adversary who knows the detection mechanism and specifically targets it.

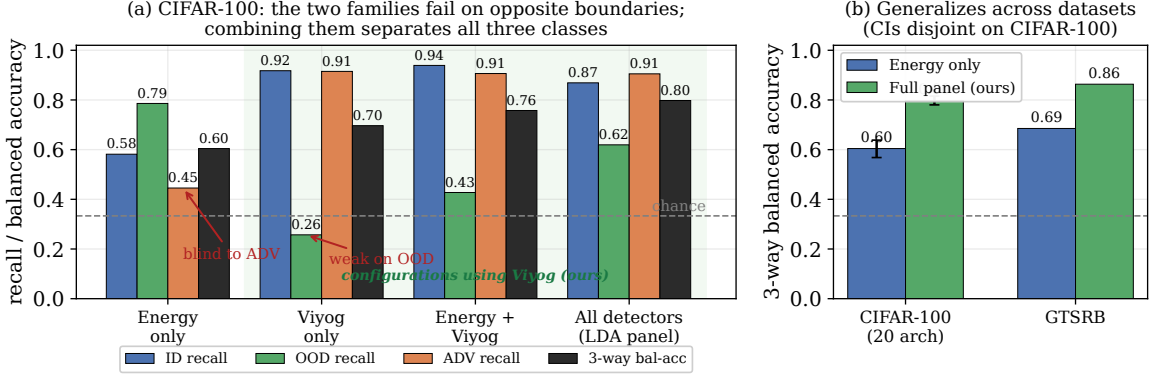


Fig. 6: Combining a logit detector with Viyog yields three-way ID/OOD/ADV separation: each covers the other’s blind spot. Logit detectors (Energy) catch low-confidence OOD but are blind to high-confidence adversaries; Viyog catches adversaries (dormant-band roughness) but is weak on smooth OOD. (a) Per-class recall (CIFAR-100, 17 archs): Energy alone 0.445 on ADV, Viyog alone 0.257 on OOD; the Energy + Viyog pair raises 3-way balanced accuracy to 0.758 and the full panel to 0.798. (b) Gain transfers to GTSRB: full panel 0.863, logit+Viyog 0.768; bootstrap 95% CIs are disjoint (full-panel [0.77, 0.83] vs. Energy-only [0.58, 0.65]). Viyog complements a logit OOD detector with no retraining or extra forward pass, at $O(C)$ state.

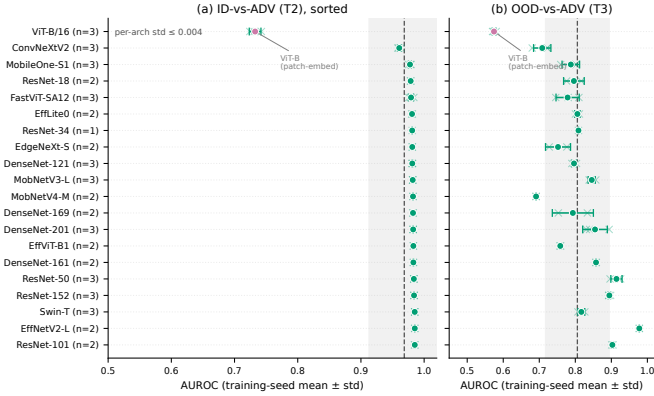


Fig. 7: Training-seed reproducibility across the full 20-architecture panel. Per-architecture mean $\pm$ std (circles), individual seeds (faint crosses), panel mean (vertical line), ± 1 overall std (band). (a) T2 seed std ≤ 0.004 across all 20 backbones, an order of magnitude below the cross-architecture spread (ViT-B the semantic-token outlier). (b) T3 similarly stable (std ≤ 0.06).

TABLE IV: Second-stage detector cost (architecture-derived). State is $O(C)$ for the first-layer detectors versus $O(D^2)/O(ND)$ for feature-distance methods. The optional “LDA head” is a tiny linear discriminant fit over the detector scores that turns the binary detector into a 3-class ID/OOD/ADV classifier ($K=3$), adding <0.1 KB and no extra forward pass.

Detector	State	Extra fwd	Order
Energy/MSP (logit)	4B	0	$O(1)$
FirstConv- L_∞ (baseline)	8B	0	$O(1)$
Viyog (primary)	$O(C)$	0	$O(C)$
+ LDA head (3-class, $K=3$)	$+<0.1$ KB	0	$O(CK)$
Mahalanobis / ViM	4.5–17 MB	1	$O(D^2)$
KNN	20–40 MB	1	$O(ND)$

Threat model. The attacker aims at both misclassification and detector evasion under L_∞ , $\varepsilon=8/255$. Fig. 13 shows

the full A0–A3 knowledge ladder; the defender’s hook, band $\mathcal{B}$, and threshold τ are fixed at deployment.

A. Adaptive Attack Formulation

A norm-aware attacker knowing that Viyog monitors first-layer L_∞ norms would optimise perturbations to preserve these norms while inducing misclassification. We maximise the following penalised PGD objective:

$$\mathcal{L}_{\text{adap}} = \mathcal{L}_{\text{CE}}(f(x+\delta), y) - \lambda (\|f_0(x+\delta)\|_\infty - \|f_0(x)\|_\infty)^2, \quad (2)$$

where $\lambda > 0$ controls the penalty strength; we sweep L_∞/L_2 /combined modes ($\varepsilon=8/255$, $\alpha=2/255$, 10 PGD steps). The reported behavior is specific to these attacks, budgets, steps, and penalty sweeps.

B. Results and Security Implications

The ladder of Fig. 13 measures what each rung of attacker knowledge costs, from black-box PGD (A0) through a both-aware adaptive adversary (A3) to a stochastic-band defence (EOT).

With the penalty active, $\|f_0(x_{\text{adap}})\|_\infty \approx \|f_0(x)\|_\infty$, keeping the raw score in-distribution. In the tested A1 configuration, the dormant-band shape read has 0% measured evasion; this observation does not follow from P1, because holding a norm fixed is not uniform rescaling. A3 reduces both evaluated signatures at a measured 8% attack-success cost.

The result holds on a second dataset. On GTSRB (ResNet-50, ConvNeXtV2, ViT-B) at max penalty λ , the both-aware attacker keeps $\geq 80\%$ success yet the combined max(dorm, HF) detector does not drop below AUROC 0.775 in these runs: single-channel evasion is backstopped by the complement (ResNet-50 dorm 0.83 $\rightarrow$ 0.51 but HF stays 0.72; ConvNeXtV2 dorm-suppression costs 80% of success). The attacker thus either keeps the detector ≥ 0.775 or surrenders most of its success, on both datasets.

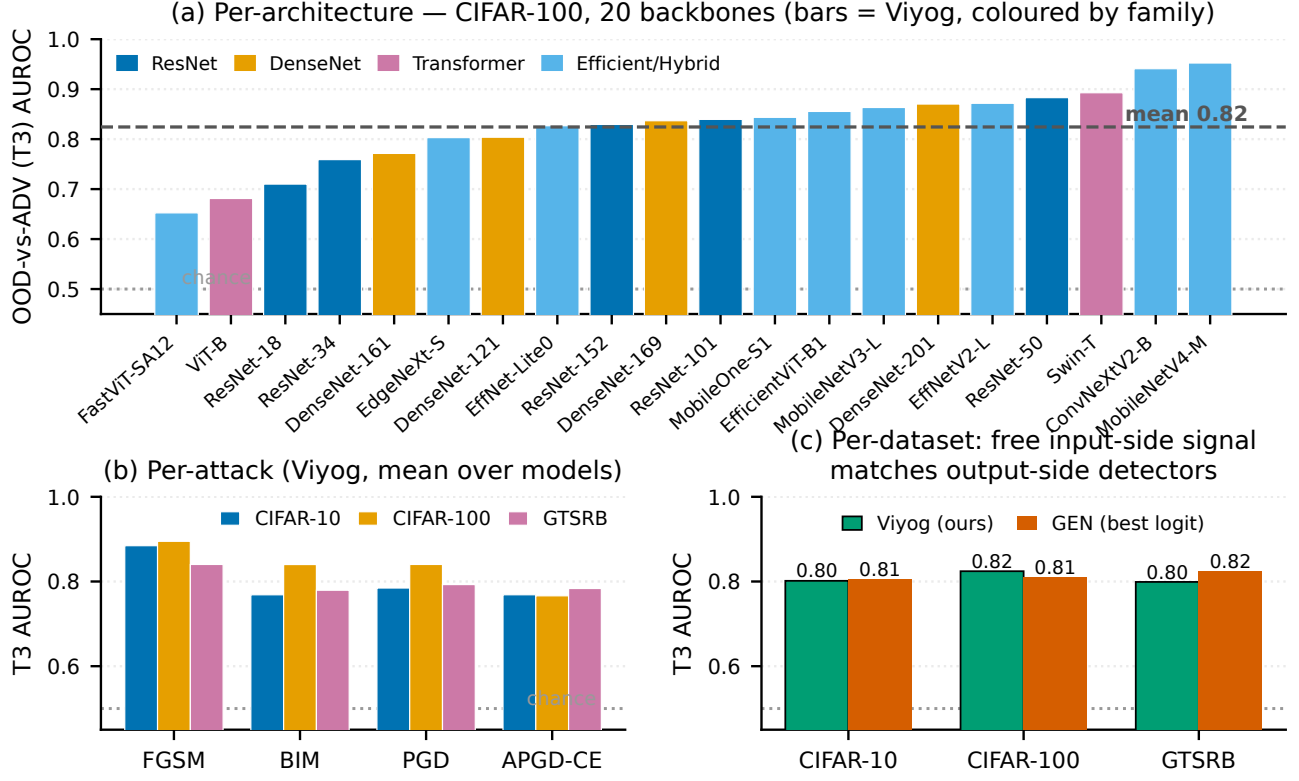


Fig. 8: The OOD-vs-ADV (T3) signal is consistent across architectures, attacks, and datasets. (a) Per-architecture T3 AUROC on 17 usable CIFAR-100 backbones (mean 0.844, range 0.68–0.95, every backbone above chance). (b) T3 across four attacks (FGSM, BIM, PGD, APGD-CE) and three datasets, weakest case ≥ 0.77 . (c) Viyog matches the strongest logit detector (GEN) within ± 0.02 at $O(C)$ state and no extra forward pass, yet stays complementary to it (Fig. 6).

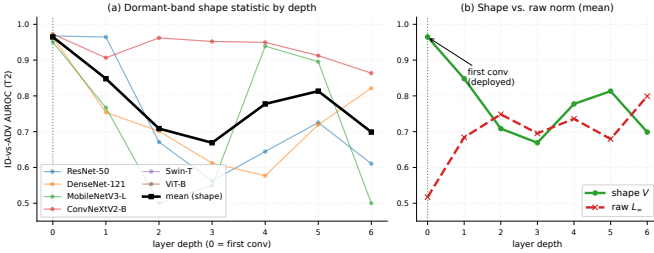


Fig. 9: The first conv is the strongest layer in the evaluated depth sweep. ID-vs-ADV (T2) AUROC of the dormant-band shape statistic V and of the raw L_∞ statistic at every depth, averaged over six architectures (ResNet-50, DenseNet-121, MobileNetV3, ConvNeXtV2, Swin-T, ViT-B) and two attacks (PGD, APGD). The shape read (green) peaks at the first conv (mean 0.965) and decays with depth, whereas the raw norm (red) is weakest there (0.52) and peaks only deep (0.80, depth 6). The first-conv shape read beats the best raw-norm layer at every evaluated depth.

Behavior of the evaluated attacks. The tested standard (A0) attacks [9]–[12], [41] naturally suppress first-layer norms and are often detected; the bespoke penalty of Eq. (2) targets that behavior directly. A different white-box optimiser may find another trade-off, so these results do not establish a universal cost of evasion [42].

Failure-mode caveat. Evading the global L_∞ score may

route a sample toward ID or OOD. Abstention is safer only if the first-stage gate still flags that sample; otherwise the adversarial can remain undetected.

C. The Robustness–Evasion Trade-Off

In the tested optimisation, norm preservation and misclassification compete, producing an 8% success cost for the both-aware attack. This is an empirical frontier for the specified optimiser, not a fundamental lower bound. A stronger held-out-basis (A4) attacker that re-optimises its perturbation basis directly against the statistic shows a similar frontier (Fig. 16), with $\sim 15\%$ of attack success surrendered in those runs. We next test whether the observed phenomenon transfers beyond the CNN architectures in the main panel.

VIII. Generalisation Beyond CNNs

The affine part of Proposition 1 applies to convolution and patch-projection pre-activations, while its empirical premises still require validation for each model. We evaluate three architecturally distinct models fine-tuned on CIFAR-100 at 224×224 resolution from ImageNet-21k pretrained weights.

ConvNeXtV2-Base [33], ViT-B/16 [34], and Swin-T [35] span depthwise-convolutional, pure self-attention, and hierarchical shifted-window designs; in every case the first

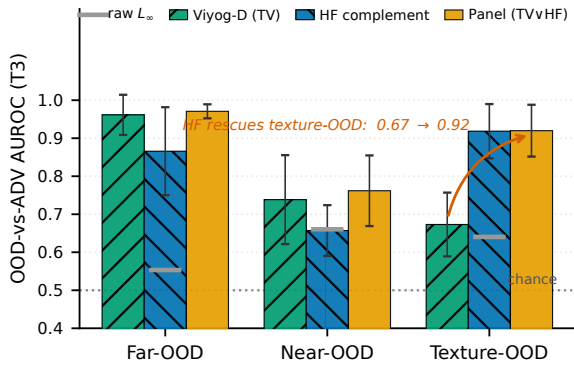
Complementary reads cover every OOD difficulty (mean $\pm$ std, 17 archs)

Fig. 10: Retrospective TV/HF diagnostic upper envelope. OOD-vs-ADV (T3) AUROC on the 17-architecture CIFAR-100 panel (mean $\pm$ std), split by OOD difficulty: far (a different dataset), near (a semantically similar shift), and texture (texture-biased). The primary dormant-band total-variation read (Viyog-D, green) is strongest on far/near-OOD but weakest on texture; its high-frequency complement (HF, blue) is the mirror image. The amber Panel takes the per-architecture better test-set statistic and reaches 0.76–0.97; it is an oracle diagnostic, not a calibrated detector. It raises texture-OOD from 0.67 to 0.92, while the raw L_∞ magnitude norm (grey) stays near chance (0.5) throughout: the separable signal is first-conv shape, not magnitude.

layer is an affine strided-convolution or patch-embedding map (ConvNeXtV2 4 $\times$ 4 conv, ViT-B 16 $\times$ 16 and Swin-T 4 $\times$ 4 patch embeddings) satisfying Eq. (3) exactly at the pre-activation stage. The implementation reads post-activation maps, for which Proposition 1 does not provide a guarantee; the GELU Taylor remainder is discussed in Appendix A.

On all three, the dormant-band roughness $V(x)$ separates ADV from ID/OOD as in the 32 $\times$ 32 CNN results, and the separation is stable over the evaluated seeds and data resamples (Fig. 7); at most three training seeds per architecture do not establish seed invariance.

These results provide evidence that the signal transfers to several convolutional and patch-projection architectures; they do not prove universal architectural generality. We next evaluate a three-way classifier.

IX. Downstream Utility: Three-Way Classification

Viyog turns a binary anomaly detector into a three-way ID/OOD/ADV classifier, an operational distinction that diagnostic metrics (AUROC, FPR) do not capture: without it a system must apply one response to all non-ID inputs, wasting adversarial purification [18]–[20] or feature-denoising [16] on benign OOD, or leaving attacks undetected. Beyond these binary routing cases, in class-incremental learning [45], [46] OOD inputs should feed back as new classes while ADV inputs must not contaminate training.

TABLE V: Two-stage recall at a disjoint-calibration 5% ID false-positive rate (mean over 17 archs $\times$ 10 split seeds).

Stage-1 gate	ADV recall	OOD recall	ID $\rightarrow$ ADV
Energy only	0.157	0.343	0.002
Energy $\oplus$ dorm-z (2-signal)	0.834	0.424	0.000

TABLE VI: Current limitations and required follow-up.

Limitation	Current boundary and required follow-up
(i) Theory	Conditional affine pre-activation bound; nonlinear hooks are empirical, with no worst-case guarantee.
(ii) Threat	Four digital L_∞ attacks at 8/255 and finite adaptive sweeps; test other norms, physical attacks, and unseen objectives.
(iii) Accuracy	T3 FPR@95 is 37.4%; cascade OOD recall is 0.424 and ViT ADV recall is 0.29; these are not stand-alone safety evidence.
(iv) Generality	Three image datasets, ≤ 3 seeds/model, one synthetic 1D task. Excludes MobileOne-S1/ResNet-18/ResNet-34 (zeroed features, Sec. V); †-marked studies (Table II) predate this and still report 20.
(v) Hardware	H200 latency is measured; edge energy is projected and requires measurement on target devices.

A. End-to-End Cascade and False-Positive Propagation

We report a two-stage analysis (non-ID gate $\rightarrow$ Viyog) over the 17-architecture panel $\times$ 10 calibration/test split seeds at a matched 5% ID-FPR (Table V, Fig. 14). Both stages’ thresholds are fit on each seed’s calibration split only and evaluated once on its disjoint test split (heldout_calibration_v1, Sec. V); this is the same protocol as the main results and excludes MobileOne-S1, ResNet-18, and ResNet-34.

The evaluated Energy gate has low adversarial recall (0.157). Adding the first-conv deviation to the gate ($\max(z_{\text{energy}}, |z_{\text{dorm}}|)$, same 5% ID-FPR) lifts it to **0.834** (mean over all 17 architectures; the weakest is ViT-B 0.29 (its patch-embedding has no dormant low-activation band), then ResNet-50 0.70, while OOD recall holds at ≈ 0.42) with mean ID $\rightarrow$ ADV mis-escalation of 0.0001 (exactly 0 on 15/17 architectures).

B. Limitations and Future Work

Table VI states the principal scientific limitations. In particular, the affine proposition is conditional and the adaptive evaluation is not a certificate.

X. Conclusion

We proposed Viyog, a gradient-free method that uses first-layer dormant-band roughness to separate OOD from ADV inputs. Under the evaluated digital L_∞ threat model, Viyog reaches T2 AUROC 0.980 across the 17-architecture held-out panel. Proposition 1 gives a sufficient pre-activation separation condition; it is not a guarantee for arbitrary attacks, natural shifts, or nonlinear hooks. In the adaptive suite, A1 has 0% measured evasion, while TV-aware attacks reduce performance.

Combined with logit scores, Viyog reaches three-way balanced accuracy 0.798 on CIFAR-100 and 0.863 on

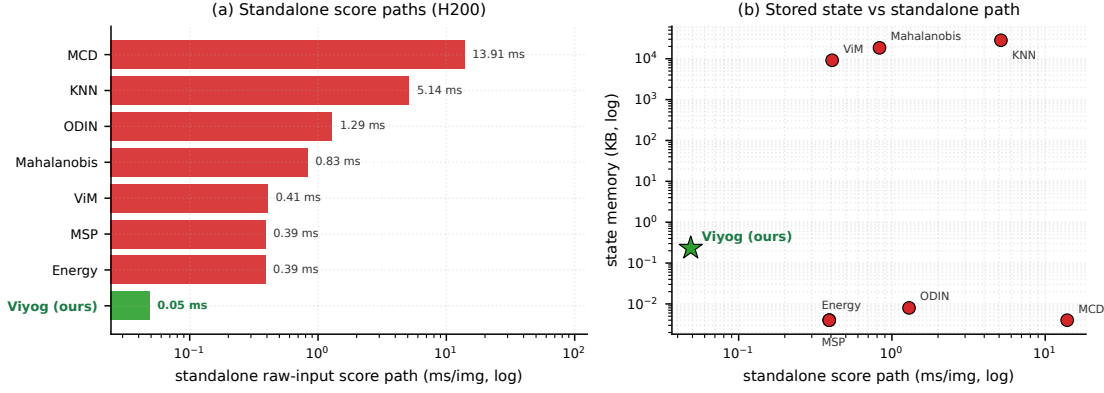


Fig. 11: Standalone score-path runtime and stored state (CIFAR-100; mean over ResNet-50, MobileNetV3, and DenseNet-121; H200 with CUDA synchronisation). (a) These raw-input timings traverse different portions of the backbone: Viyog stops after the first layer, whereas logit and deep-feature scores require later layers. They therefore describe standalone score paths, not incremental second-stage latency or deployed speedup. The Viyog path is 0.05 ms/img. (b) The plotted state comparison shows 0.23 KB for the plotted 64-channel profile against the 9–28 MB covariance/feature-bank of the distance detectors (10^4 – $10^5 \times$ smaller). Incremental overhead for a normal full inference is reported separately in Fig. 12.

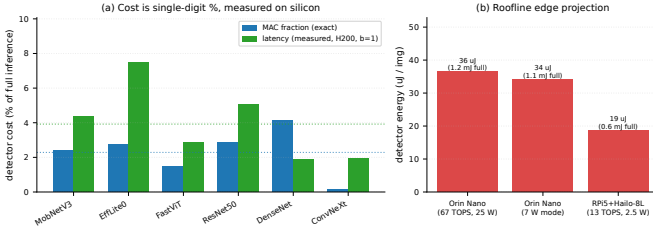


Fig. 12: Measured H200 detector cost and edge projection. (a) Per-architecture cost of the primary $V(x)$ detector as a fraction of full inference: the exact MAC fraction (mean 2.28%) and the measured batch-1 latency on an H200 (CUDA events; mean 3.9%) show a single-digit-% second-stage tax on this platform silicon across CNNs, a ViT, and a ConvNeXt. (b) Compute-bound roofline projection of the detector’s per-image energy on real edge accelerators (Jetson Orin Nano, RPi5 + Hailo-8L): 19–37 μ J versus a 0.6–1.2 mJ full inference. These compute-bound figures are optimistic lower bounds; memory, runtime, and power effects require measurement on each target device.

GTSRB, adding no learned parameters and $O(C)$ state at 2.28% CNN MAC / 3.9% H200 latency overhead. High T3 FPR, weak cases, limited adaptive coverage, and projected edge energy prevent stand-alone deployment claims without the follow-up in Table VI.

Appendix A

General-Affine Bound, GELU Remainder, and Proof of Proposition 1

The first-layer perturbation bound (Eq. (3)) uses only that the first layer is affine. For any first-layer Jacobian W (dense layer, strided convolution, or ViT/Swin patch embedding) and perturbation $\|\delta\|_\infty \leq \varepsilon$,

$$\|W\delta\|_\infty \leq \|W\|_\infty \varepsilon, \quad \|W\|_\infty = \max_i \sum_j |W_{ij}|, \quad (3)$$

the induced ∞ -operator norm (maximum absolute row sum). Convolution is the special case in which $W = M_m$ is Toeplitz and the maximum row sum equals $\|W_m\|_1$, recovering Eq. (3); the Toeplitz structure is thus sufficient but not necessary, so the mechanism extends to patch-embedding first layers that have no shift-invariance.

If the first activation is a smooth g (e.g. GELU), the linearisation error in each pre-activation coordinate is bounded by the Taylor remainder $\frac{1}{2} \sup_z |g''(z)| (\|W\|_\infty \varepsilon)^2$. For $\text{GELU}(z) = z \Phi(z)$, $g''(z) = \varphi(z)(2 - z^2)$ has interior extrema at $z = 0$ (value $2\varphi(0)$) and $z = \pm 2$ (value $-2\varphi(2) \approx -0.108$); since $0.798 > 0.108$ the supremum is at $z = 0$,

$$\sup_z |\text{GELU}''(z)| = 2\varphi(0) = \sqrt{2/\pi} \approx 0.798, \quad (4)$$

so the per-coordinate error is at most $0.399 (\|W\|_\infty \varepsilon)^2$. Normalising this quadratic bound by the perturbation envelope $\|W\|_\infty \varepsilon$ gives $0.399 \|W\|_\infty \varepsilon$, which is $O(1)$ – $O(10)$ for the measured first-layer weights (e.g. ≈ 6.8 for ResNet-50 at $\varepsilon = 8/255$). This is not a relative-error guarantee because the actual linear response may be smaller than its envelope. The calculation therefore does not certify a post-GELU version of Proposition 1; it only quantifies a worst-case local remainder.

Full Proof of Proposition 1: The affine residue $e_c = w_c * \delta$ established above is the object on which the roughness statistic acts; we now prove its three properties (P1–P3) in turn.

(P1) Uniform rescaling. Absolute 1-homogeneity gives

$$\widetilde{\text{TV}}(\lambda_c a_c) = \frac{\lambda_c \text{TV}(a_c)}{\lambda_c \nu_c + \epsilon} = \frac{\text{TV}(a_c)}{\nu_c + \epsilon/\lambda_c}.$$

This is exact invariance only when $\epsilon = 0$; it makes no claim about perturbations that preserve a norm while changing spatial structure.

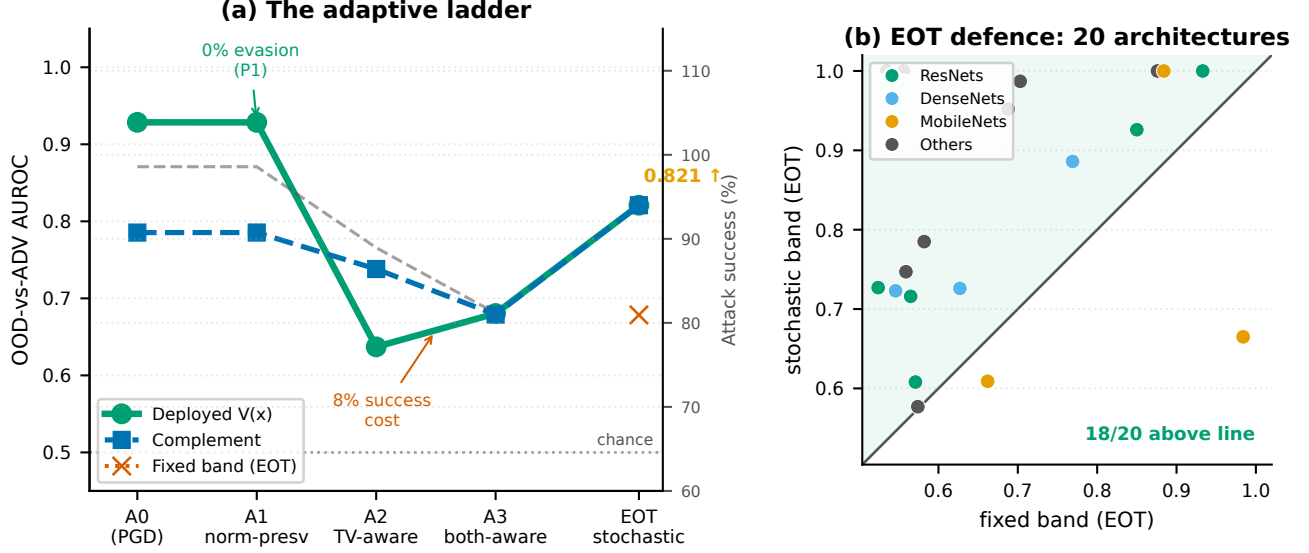


Fig. 13: Defence-in-depth against an escalating adaptive adversary (mean over 20 architectures). Attacker knowledge grows left-to-right: A0 (PGD), A1 norm-preserving, A2 TV-aware (white-box), A3 both-aware, EOT stochastic defence. (a) A1 achieves 0% evasion empirically; A2 erodes V to 0.64 while the complement holds at 0.74; EOT restores V to 0.82; A3 suppresses both only at an 8% attack-success cost. (b) EOT scatter: 18/20 architectures are above the diagonal; this supports stochastic band-randomisation in the tested setting but is not a robustness certificate.

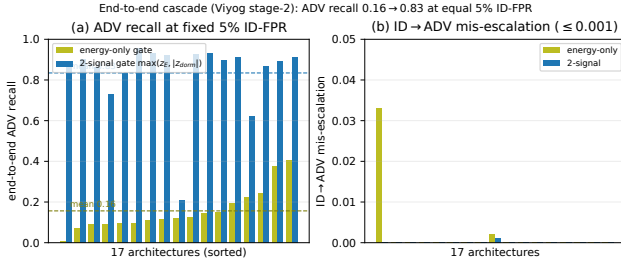


Fig. 14: Two-stage cascade over the 17-architecture panel at a disjoint-calibration 5% ID-FPR. (a) Energy-only gate: mean ADV recall 0.157; two-signal gate ($\max(z_{\text{energy}}, |z_{\text{dorm}}|)$, where z_X denotes the normalised score for statistic X): mean 0.834. (b) ID→ADV mis-escalation is ≤ 0.0001 .

(P2) Two-sided bound. Affineness gives $e_c = w_c * \delta$ exactly. TV subadditivity and reverse triangle give

$$[\rho_c r_c - \beta_c] + \nu_c \leq \text{TV}(a_c + e_c) \leq (\beta_c + \rho_c r_c) \nu_c.$$

The spatial L_1 triangle and reverse-triangle inequalities give

$$\nu_c |1 - r_c| \leq \overline{|a_c + e_c|} \leq \nu_c (1 + r_c).$$

For the lower ratio bound, divide the lower numerator by the upper denominator; for the upper ratio bound, divide the upper numerator by the lower denominator. Adding $\epsilon = \eta_c \nu_c$ and cancelling ν_c yields P2.

(P3) Conditional separation. A natural comparison satisfying A1 has normalised TV at most β_c . Hence P2 gives

$$\widetilde{\text{TV}}(a'_c) - \widetilde{\text{TV}}(a_c) \geq \frac{\rho_c r_c - \beta_c (2 + r_c + \eta_c)}{1 + r_c + \eta_c}.$$

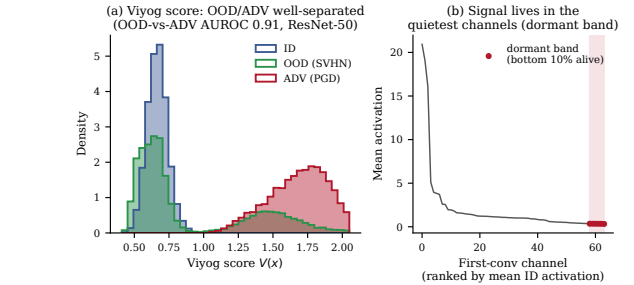


Fig. 15: The dormant-band signature. Per-image dormant-band TV profile for ID (green), OOD (blue), and adversarial (red) inputs on ResNet-50 (CIFAR-100, five OOD sets, PGD attack). ID and OOD share a low, smooth profile; adversaries inject high-frequency structure, separating at AUROC 0.91.

The stated condition makes each margin positive, and averaging over the fixed band $\mathcal{B}$ yields the claim. The result is conditional on A1–A3 and does not extend automatically through ReLU or GELU.

Appendix B

Non-Vision Sanity Check (1D Signals)

As a preliminary check beyond images, we evaluate a 1D-CNN trained on band-limited multi-sinusoid signals. It detects adversarials at mean T2 AUROC **0.96** (vs. 0.71 for the raw norm, 3 seeds; Fig. 17). This synthetic experiment is evidence for one 1D setting, not a claim of modality-agnostic performance.

Artifact Availability

The evaluated MIT-licensed source, tests, paper source, and reproduction entry point are archived as release

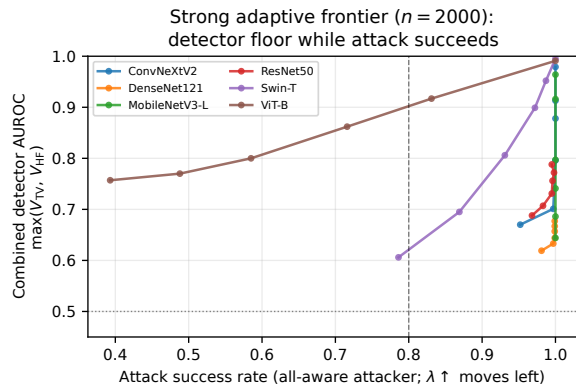


Fig. 16: Strong adaptive frontier: held-out basis (6 archs). An A4 attacker that re-optimises its perturbation basis directly. Dorm/HF mean 0.86/0.79 (min 0.54/0.53); only the both-aware objective suppresses both to ≈ 0.62 at a 15% attack-success cost in this evaluated configuration.

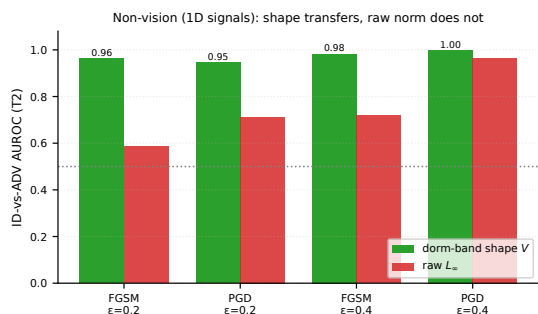


Fig. 17: Synthetic 1D sanity check. ID-vs-ADV AUROC of $V(x)$ (green) vs. raw L_∞ (red) at the first Conv1d layer for FGSM/PGD at two budgets ($n=1,500$, 3 seeds). Shape reaches 0.95–1.00 while the raw norm stays near chance; this single synthetic task is preliminary evidence, not a general non-vision validation.

v0.1.3-codes26-ae-6 at Zenodo [47] and mirrored by the exact GitHub release. The detector is distributed as `viyog==0.1.3`, and the checkpoints are pinned at Hugging Face. The canonical entry point is `reproduce_t1.sh`; the precomputed adversarial dataset and live Space linked from the archive README are optional and are not required by that script.

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